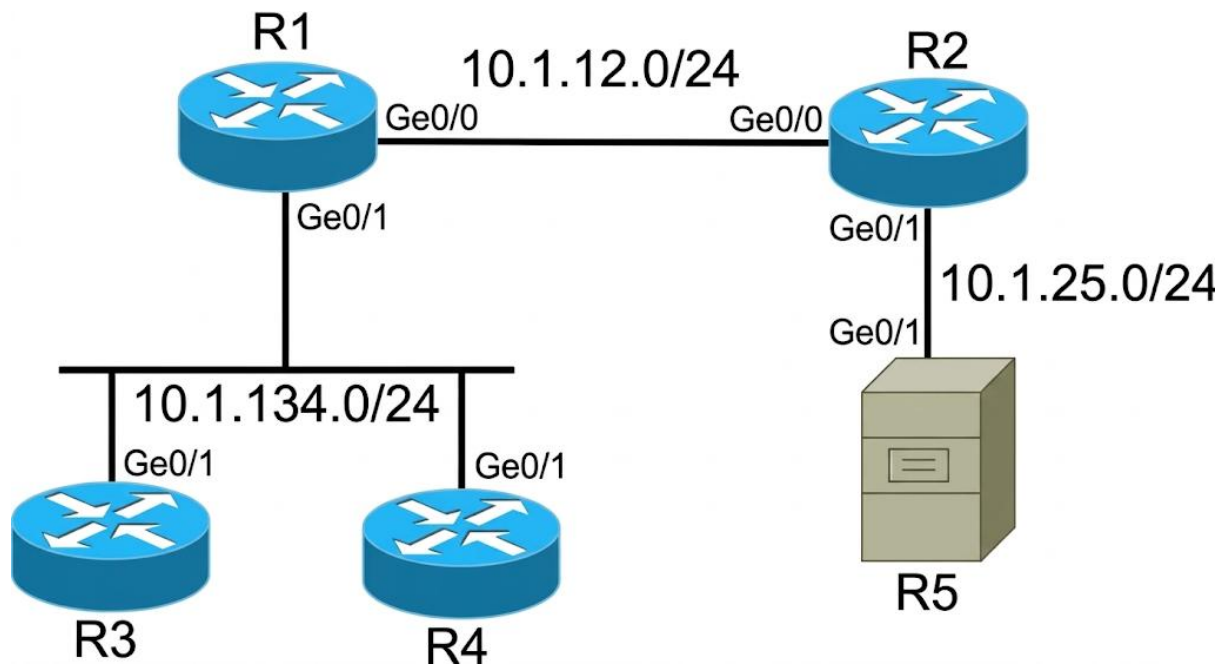


Port Address Translation (PAT)



Objective

Configure Port Address Translation (PAT) / NAT Overload using both a single-address NAT pool and direct interface overloading.

Tasks

- Configure hostnames and IP addresses across all devices.
- Disable routing on **R3** and **R4** (`default-gateway 10.1.134.1`).
- Disable routing on **R5** (`default-gateway 10.1.25.2`).
- Configure default routing on **R1** via `10.1.12.2`.
- Enable HTTP service on **R5**.
- Configure PAT for `10.1.134.0/24` using Option 1 (Single-Address Pool).
- Verify end-to-end traffic from **R3** and **R4** to **R5**.
- Reconfigure PAT using Option 2 (Direct Interface Overload) and re-verify.

Concept & Overview

Port Address Translation (PAT), or NAT Overload, extends basic NAT by multiplexing thousands of internal private IP flows onto a single public IP address (or small pool of addresses). PAT tracks

connections by appending unique layer-4 source port numbers to the translated public IP, supporting up to ~64,000 concurrent sessions per IP address.

PAT can be configured via two primary techniques:

1. **Pool-based Overload:** Useful when assigning a dedicated public address distinct from interface addresses.
2. **Interface-based Overload:** Ideal when the outside public IP is dynamically assigned via DHCP or IPCP.

Device Configurations

Router R1 – Option 1: Single-Address Pool

```
enable
```

```
configure terminal
```

```
hostname R1
```

```
!
```

```
interface GigabitEthernet0/1
```

```
ip address 10.1.134.1 255.255.255.0
```

```
ip nat inside
```

```
!
```

```
interface GigabitEthernet0/0
```

```
ip address 10.1.12.1 255.255.255.0
```

```
ip nat outside
```

```
!
```

```
ip route 0.0.0.0 0.0.0.0 10.1.12.2
```

```
!  
  
access-list 1 permit 10.1.134.0 0.0.0.255  
  
!  
  
ip nat pool TRANSLATE 10.1.12.3 10.1.12.3 prefix-length 24  
  
ip nat inside source list 1 pool TRANSLATE overload  
  
!  
  
end
```

(Configurations for R2, R3, R4, and R5 match previous labs).

Verification – Option 1

Execute outbound connections from **R3** and **R4**:
R3#ping 10.1.25.5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.1.25.5, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 84/100/116 ms

R4#telnet 10.1.25.5

Trying 10.1.25.5 ... Open

User Access Verification Password:

R5>

Inspect translation entries on **R1**:
R1#show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
-----	---------------	--------------	---------------	----------------

```
icmp 10.1.12.3:1024 10.1.134.3:18 10.1.25.5:18 10.1.25.5:1024
```

```
tcp 10.1.12.3:4096 10.1.134.4:65284 10.1.25.5:23 10.1.25.5:23
```

Both host IP addresses (**10.1.134.3** and **10.1.134.4**) translate to the single public pool IP (**10.1.12.3**), differentiated by source ports **1024** and **4096**.

Configuration & Verification – Option 2: Direct Interface Overload

Remove the address pool configuration from **R1** and bind translation directly to interface **GigabitEthernet0/0**:
R1(config)# no ip nat inside source list 1 pool TRANSLATE overload

```
R1(config)# no ip nat pool TRANSLATE 10.1.12.3 10.1.12.3 prefix-length 24
```

```
!
```

```
R1(config)# ip nat inside source list 1 interface GigabitEthernet0/0 overload
```

```
!
```

```
end
```

Generate new traffic from **R3** and **R4**:

```
R3#ping 10.1.25.5
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.1.25.5, timeout is 2 seconds:

```
!!!!
```

Success rate is 80 percent (4/5), round-trip min/avg/max = 44/79/100 ms

```
R4#telnet 10.1.25.5
```

Trying 10.1.25.5 ... Open

User Access Verification Password:

```
R5>
```

Verify the updated translation table:

```text

R1#show ip nat translations

| Pro  | Inside global  | Inside local     | Outside local | Outside global |
|------|----------------|------------------|---------------|----------------|
| icmp | 10.1.12.1:1024 | 10.1.134.3:20    | 10.1.25.5:20  | 10.1.25.5:1024 |
| tcp  | 10.1.12.1:4096 | 10.1.134.4:65528 | 10.1.25.5:23  | 10.1.25.5:23   |

Without an IP pool, both internal hosts translate dynamically using interface **GigabitEthernet0/0**'s IP address (**10.1.12.1**), conserving IPv4 space efficiently without dedicated pool management.

## Summary Comparison

| NAT Type       | Configuration Mode                                                    | Translation Ratio |          |
|----------------|-----------------------------------------------------------------------|-------------------|----------|
| Static NAT     | <code>ip nat inside source static [local] [global]</code>             | 1-to-1 Mapping    | Hi       |
| Dynamic NAT    | <code>ip nat inside source list [ACL] pool [POOL]</code>              | 1-to-1 Dynamic    | M<br>po  |
| PAT (Overload) | <code>ip nat inside source list [ACL] interface [INT] overload</code> | Many-to-1 Mapping | Lo<br>su |